

MATH3018 Assignment 6, due on Apr 7

- (1) Let $X = (X_1, X_2, X_3)$ have a population covariance matrix

$$\Sigma = \begin{pmatrix} \sigma^2 & \sigma^2\rho & 0 \\ \sigma^2\rho & \sigma^2 & \sigma^2\rho \\ 0 & \sigma^2\rho & \sigma^2 \end{pmatrix}, -\frac{1}{\sqrt{2}} < \rho < \frac{1}{\sqrt{2}}$$

Find the principal components and the variance explained by each.

- (2) $n = 10$ observations from a bivariate population are listed as follows.

Observation #	X_1	X_2
1	108.28	17.05
2	152.36	16.59
3	95.04	10.91
4	65.45	14.14
5	62.97	9.52
6	263.99	25.33
7	265.19	18.54
8	285.06	15.73
9	92.01	8.10
10	165.68	11.13

- (a) Using sample covariance matrix, determine the sample principal components Y_1 and Y_2 , and their variances.
 - (b) Find the proportion of the total sample variance explained by Y_1 .
 - (c) Compute the correlation coefficients $\text{Cor}(Y_1, X_j)$, for $j = 1$ and 2 respectively.
 - (d) Let $Z = (Z_1, Z_2)^T$ denote the standardized variables. Determine the sample principal components Y_1^Z and Y_2^Z , and their variances.
 - (e) Find the proportion of the total sample variance explained by Y_1^Z .
 - (f) Compute the correlation coefficients $\text{Cor}(Y_1^Z, Z_j)$ for $j = 1$ and 2 respectively. Comment on Y_1^Z .
 - (g) Comment on your findings comparing Questions (2) and (3).
- (3) The weekly rates of return for five stocks listed on the New York Stock Exchange are given in Table 1 below. JPMorgan, CitiBank and WellsFargo are financial service companies; RoyDutShell and ExxonMobile are energy companies. (the full dataset “stocks-dat” has been uploaded to Moodle).
- (a) Construct the sample covariance matrix S , and find the sample principal components.
 - (b) Determine the proportion of the total sample variance explained by the first three principal components. Interpret these components.
 - (c) Construct Bonferroni simultaneous 90% confidence intervals for the variances λ_1, λ_2 , and λ_3 of the first three population components Y_1, Y_2 , and Y_3 .
 - (d) Given the results in Parts (a)-(c), do you feel that the stock rates-of-return data can be summarized in fewer than five dimensions? Explain.

TABLE 1. Stock-price data (weekly rate of return)

Week	JPMorgan	CitiBank	WellsFargo	RoyDutShell	ExxonMobil
1	0.0130338	-0.0078431	-0.0031889	-0.0447693	0.0052151
2	0.0084862	0.0166886	-0.0062100	0.0119560	0.0134890
3	-0.0179153	-0.0086393	0.0100360	0.0000000	-0.0061428
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
102	0.0103929	-0.0026612	0.0044290	-0.0024818	-0.0164502
103	-0.0127948	-0.0143678	-0.0187402	-0.0049759	-0.0163732